

Small-Noise Mesh Laws and the Double-Pole Obstruction for Intrinsic Bulk Two-Step Fredholm Determinants

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July 2026

Abstract

At every fixed positive noise width, the Perron/parity-deflated folded-Gaussian operator has an intrinsic trace-class square and a strict Fredholm determinant. The next problem contains two logically independent limits: spatial resolution $n \rightarrow \infty$ and noise width $\sigma \downarrow 0$. We separate them.

First, for $0 < \sigma \leq 0.03$, explicit Gaussian-moment estimates give

$$\|\mathcal{K}_\sigma\|_{\mathfrak{S}_2} \leq A_0 \sigma^{-1/2}, \quad \|(k_\sigma)_x\|_{L^2} + \|(k_\sigma)_y\|_{L^2} \leq A_1 \sigma^{-3/2},$$

with $A_0 = 2.124503864054395$ and $A_1 = 11.373709849581182$. If $G_{n,\sigma}$ is the orthogonal cell-average Galerkin lift, then

$$\|G_{n,\sigma} - \mathcal{K}_\sigma\|_{\mathfrak{S}_2} \leq \frac{A_1}{\pi n \sigma^{3/2}},$$

and

$$\|G_{n,\sigma}^2 - \mathcal{K}_\sigma^2\|_{\mathfrak{S}_1} \leq \frac{15.38295583703884}{n \sigma^2} + \frac{13.10703757570889}{n^2 \sigma^3}.$$

Thus $n \sigma^{3/2} \rightarrow \infty$ is sufficient for one-step Hilbert–Schmidt convergence, while $n \sigma^2 \rightarrow \infty$ is sufficient for two-step trace-norm convergence. A normalized Gaussian row model proves that the exponent $h \sigma^{-3/2}$ is sharp for cell projection, with exact constant $(\sqrt{48} \pi^{1/4})^{-1}$. On shrinking determinant disks $|w| \leq \rho \sigma$, the standard continuity estimate only requires $n \sigma \rightarrow \infty$; on a fixed disk it develops an exponential $e^{C_R/\sigma}$ wall.

Second, arbitrarily fine spatial resolution cannot produce a naive entire small-noise limit. The deterministic parity-centered one-step germ is known exactly in the form

$$\widehat{D}_{0,\text{bulk},2}(z) = \frac{\mathcal{G}(z)}{1 - z^2/\lambda}, \quad \lambda = 1.678573510428322 \dots,$$

where $\mathcal{G}$ is holomorphic and nonzero for $|z| < \lambda$. Symmetric regularization therefore gives the two-step germ

$$\widehat{\mathcal{F}}_0(w) = \frac{\mathcal{H}(w)}{(1 - w/\lambda)^2},$$

where $\mathcal{H}$ is holomorphic and nonzero for $|w| < \lambda^2$. Hence $w = \lambda$ is a genuine double pole. Although every positive-noise two-step determinant is entire and its Taylor coefficients converge, the family is not locally bounded on any disk of radius greater than λ .

The canonical finite resolution of the pole is

$$S_N(w) = \Pi_N(w/\lambda)^2, \quad \Pi_N(q) = 1 + q + \dots + q^N,$$

and its edge profile is

$$\frac{S_N(\lambda e^{s/(N+1)})}{(N+1)^2} \longrightarrow \left(\frac{e^s - 1}{s} \right)^2.$$

Archived resonance clouds exhibit this squared profile numerically. The remaining open gate is a uniform small-noise weighted-Riesz transport theorem; it is stated explicitly rather than assumed silently. No arithmetic trace formula, self-adjoint Hilbert–Pólya operator, or Riemann-hypothesis claim is made.

Keywords: small noise; Hilbert–Schmidt approximation; trace norm; Fredholm determinant; normal family; resonance cloud; Gaussian kernel.

MSC 2020: 47B10; 47A10; 65R20; 37M25; 30D45.

1 Introduction

The fixed-noise spectral construction has reached a strict determinant. For the folded-Gaussian Markov operator $\mathcal{K}_\sigma$, the Perron and negative parity weighted Riesz terms can be removed intrinsically at $\sigma = 10^{-2}$, giving

$$\mathcal{B}_\sigma = \mathcal{K}_\sigma - \mathcal{Q}_{+,\sigma} - \mathcal{Q}_{-,\sigma}.$$

The preceding trace-ideal completion proved that full and adaptive discretizations converge to $\mathcal{B}_{10^{-2}}$ in Hilbert–Schmidt norm, their squares converge in trace norm, and

$$\det(\mathbf{I} - wB_{n,10^{-2}}^2) \longrightarrow \det(\mathbf{I} - w\mathcal{B}_{10^{-2}}^2) \quad (1)$$

locally uniformly in w [6].

It is tempting to let σ decrease and merely increase the matrix dimension. This conflates two different obstructions.

1. The Gaussian kernel becomes narrower and larger in Hilbert–Schmidt norm. A grid schedule adequate for eigenvalue pictures need not make the lifted kernel converge in an absolute trace ideal.
2. Even at infinite spatial resolution, the deterministic parity-centered target is meromorphic, not entire. No mesh schedule can remove a genuine pole in the small-noise coefficient germ.

The purpose of this paper is to separate those effects quantitatively. The first half derives an unconditional mesh law for the raw folded Gaussian operator. Its one-step scale is $n\sigma^{3/2}$, not the commonly used $n\sigma$. Squaring introduces the growing Hilbert–Schmidt size $\|\mathcal{K}_\sigma\|_2 = O(\sigma^{-1/2})$ and moves the sufficient trace-norm scale to $n\sigma^2$.

The second half combines the fixed-noise symmetric determinant identity with the exact deterministic pole theorem of the parity-extracted bulk analysis [3]. The simple poles at $z = \pm\sqrt{\lambda}$ merge under $w = z^2$ into a double pole at $w = \lambda$. This gives a stronger conclusion than “the numerical limit looks unstable”: the family cannot be locally bounded on any disk crossing that point.

The correct finite model is also forced. A simple one-step pole is resolved by the geometric polynomial $\Pi_N(z^2/\lambda)$. The two-step symmetric product squares this factor. The resonance cloud therefore produces double zeros in the w -plane and the universal edge profile is the square of the one-step scattering profile.

Logical layers

Four levels are kept distinct throughout.

Unconditional analytic theorem

Uniform Gaussian Hilbert–Schmidt bounds, cell-average Galerkin mesh laws, the sharp normalized-row exponent, the deterministic double pole, the normal-family obstruction, and the exact geometric square model.

Fixed-noise validated input

The $\sigma = 10^{-2}$ intrinsic weighted-Riesz terms and strict bulk-square determinant from RH-45.

Conditional bulk route

A uniform small-noise bound for the peripheral weighted terms and their discretizations. This is isolated as a named condition.

Floating diagnostic

Reprocessing of the archived RH-15 resonance clouds in the two-step variable. It supports the geometric mechanism but proves no uniform contour theorem.

2 Folded Gaussian family and cell lifts

Let u_c be the root in (1.5, 1.6) of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (2)$$

and put $m(x) = 1 - u_c x^2$ for $x \in [0, 1]$. For $0 < \sigma \leq 0.03$, define

$$g_\sigma(x, y) = e^{-(y-m(x))^2/(2\sigma^2)} + e^{-(y+m(x))^2/(2\sigma^2)}, \quad (3)$$

$$Z_\sigma(x) = \int_0^1 g_\sigma(x, y) dy, \quad k_\sigma(x, y) = \frac{g_\sigma(x, y)}{Z_\sigma(x)}, \quad (4)$$

$$(\mathcal{K}_\sigma f)(x) = \int_0^1 k_\sigma(x, y) f(y) dy. \quad (5)$$

The omitted factor $(\sqrt{2\pi}\sigma)^{-1}$ cancels in the row normalization.

Let V_n be the space of functions constant on the cells $I_{n,j} = [j/n, (j+1)/n)$, and let E_n be the orthogonal projection onto V_n . The canonical cell-average Galerkin operator is

$$G_{n,\sigma} = E_n \mathcal{K}_\sigma E_n, \quad (6)$$

extended by zero on $V_n^\perp$. The Euclidean matrix in the normalized cell basis is isometric to this lift, so matrix Frobenius norm equals operator Hilbert–Schmidt norm.

The distinction between (6) and a midpoint matrix matters only for constants. Cell averaging makes the leading small-noise exponent transparent and gives an exact orthogonal projection. Midpoint sampling, row normalization, and adaptive cutoff are discussed after the leading law has been established.

3 Uniform Gaussian Hilbert–Schmidt envelope

The folded normalizer has a useful exact interpretation. With $t = -y$ in the second term of (3),

$$Z_\sigma(x) = \int_{-1}^1 e^{-(t-m(x))^2/(2\sigma^2)} dt. \quad (7)$$

Since $m(x) \in [1 - u_c, 1]$ and $|1 - u_c| < 1$, the minimum occurs at $m = 1$. Therefore

$$Z_\sigma(x) \geq c_0 \sigma, \quad c_0 = \sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{\sqrt{2}}{0.03}\right) = 1.2533141373155 \dots \quad (8)$$

Lemma 3.1 (Raw Gaussian row bounds). *Uniformly in x and $0 < \sigma \leq 0.03$,*

$$\|g_\sigma(x, \cdot)\|_{L^2(0,1)} \leq 2\pi^{1/4} \sigma^{1/2}, \quad (9)$$

$$\|(g_\sigma)_y(x, \cdot)\|_{L^2(0,1)} \leq \sqrt{2} \pi^{1/4} \sigma^{-1/2}, \quad (10)$$

$$\|(g_\sigma)_m(x, \cdot)\|_{L^2(0,1)} \leq \sqrt{2} \pi^{1/4} \sigma^{-1/2}, \quad (11)$$

$$|(Z_\sigma)_m(x)| \leq 1. \quad (12)$$

Proof. Use $(a + b)^2 \leq 2(a^2 + b^2)$ and enlarge each target integral to the real line. The required moments are

$$\int_{\mathbb{R}} e^{-t^2/\sigma^2} dt = \sqrt{\pi}\sigma, \quad \int_{\mathbb{R}} \frac{t^2}{\sigma^4} e^{-t^2/\sigma^2} dt = \frac{\sqrt{\pi}}{2\sigma}.$$

Differentiating (7) in the center gives a difference of two endpoint exponentials, both in $[0, 1]$, proving (12). $\square$

Theorem 3.2 (Explicit small-noise envelope). *For $0 < \sigma \leq 0.03$,*

$$\|\mathcal{K}_\sigma\|_{\mathfrak{S}_2} \leq A_0 \sigma^{-1/2}, \quad A_0 = \frac{2\pi^{1/4}}{c_0} \leq 2.124503864054395, \quad (13)$$

$$\|(k_\sigma)_x\|_{L^2([0,1]^2)} + \|(k_\sigma)_y\|_{L^2([0,1]^2)} \leq A_1 \sigma^{-3/2}, \quad A_1 \leq 11.373709849581182. \quad (14)$$

Proof. Equation (13) follows from (8) and (9). The target derivative follows from (10). For the center derivative,

$$(k_\sigma)_m = \frac{(g_\sigma)_m}{Z_\sigma} - \frac{g_\sigma(Z_\sigma)_m}{Z_\sigma^2}, \quad (15)$$

so lemma 3.1 gives

$$\|(k_\sigma)_m(x, \cdot)\|_2 \leq \pi^{1/4} \left(\frac{\sqrt{2}}{c_0} + \frac{2}{c_0^2} \right) \sigma^{-3/2}. \quad (16)$$

Finally, $|m'(x)| \leq 2u_c$. Adding the source and target constants yields A_1 . $\square$

The estimates are deliberately coarse. At $\sigma = 10^{-2}$ they give

$$\|\mathcal{K}_\sigma\|_2 \leq 21.2451, \quad \|k_x\|_2 + \|k_y\|_2 \leq 11373.8,$$

whereas the 160-bit fixed-noise certificate gives 5.49855 and 1052.04. The present constants are larger because they remain valid uniformly down to zero noise without interval integration at every width.

4 The sharp Gaussian-row projection exponent

The power $\sigma^{-3/2}$ is not merely an artifact of the upper bound. Consider the normalized Gaussian density on $\mathbb{R}$,

$$f_\sigma(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-y^2/(2\sigma^2)}, \quad (17)$$

and let P_h denote orthogonal averaging on cells of width h .

Theorem 4.1 (Sharp cell-average row law). *As $h/\sigma \rightarrow 0$,*

$$\|f_\sigma - P_h f_\sigma\|_{L^2(\mathbb{R})} \sim C_G h \sigma^{-3/2}, \quad C_G = \frac{1}{\sqrt{48} \pi^{1/4}} = 0.1084156338230097 \dots \quad (18)$$

The law is independent of the phase of the cell grid relative to the Gaussian center.

Proof. The standard cell-average expansion gives

$$\|f - P_h f\|_2^2 = \frac{h^2}{12} \|f'\|_2^2 + o(h^2) \quad (19)$$

for $f \in H^1(\mathbb{R})$; translation of the grid changes only the lower-order term. Direct Gaussian integration gives

$$\|f'_\sigma\|_2^2 = \frac{1}{4\sqrt{\pi}\sigma^3}.$$

Insert this in (19). $\square$

There is also an exact scaling identity. If $a = h/\sigma$, then

$$\|f_\sigma - P_h f_\sigma\|_2 = \sigma^{-1/2} E(a), \quad E(a) \sim C_G a. \quad (20)$$

The numerical pilot evaluates every projected cell mass by differences of the normal distribution function. At $a = 2^{-9}$, the scaled constant is within 9.49×10^{-8} relative error of C_G .

The row theorem does not by itself give a lower bound for the complete folded operator, because source averaging and boundary conditioning add geometry. It does show that the exponent in (14) is the natural absolute- L^2 cell-resolution exponent for a Gaussian row.

5 One-step and two-step mesh laws

The tensor Poincaré–Wirtinger inequality and [theorem 3.2](#) immediately give the leading grid law.

Theorem 5.1 (Explicit cell-average mesh bounds). *For every $n \geq 2$ and $0 < \sigma \leq 0.03$,*

$$\varepsilon_{n,\sigma} := \|G_{n,\sigma} - \mathcal{K}_\sigma\|_{\mathfrak{S}_2} \leq \frac{A_1}{\pi n \sigma^{3/2}} \leq \frac{3.620364287707642}{n \sigma^{3/2}}. \quad (21)$$

Moreover,

$$\begin{aligned} \delta_{n,\sigma} &:= \|G_{n,\sigma}^2 - \mathcal{K}_\sigma^2\|_{\mathfrak{S}_1} \\ &\leq \varepsilon_{n,\sigma} (2A_0 \sigma^{-1/2} + \varepsilon_{n,\sigma}) \end{aligned} \quad (22)$$

$$\leq \frac{15.38295583703884}{n \sigma^2} + \frac{13.10703757570889}{n^2 \sigma^3}. \quad (23)$$

Proof. The kernel of $E_n \mathcal{K}_\sigma E_n$ is the conditional expectation of k_σ on the n^2 cells. Applying the one-dimensional cellwise Poincaré inequality successively in x and y gives

$$\|\mathcal{K}_\sigma - E_n \mathcal{K}_\sigma E_n\|_2 \leq \frac{1}{\pi n} (\|k_x\|_2 + \|k_y\|_2).$$

This proves (21). Products of two Hilbert–Schmidt operators are trace class, so

$$\|G_{n,\sigma}^2 - \mathcal{K}_\sigma^2\|_1 \leq \|G_{n,\sigma} - \mathcal{K}_\sigma\|_2 (\|G_{n,\sigma}\|_2 + \|\mathcal{K}_\sigma\|_2).$$

Use $\|G_{n,\sigma}\|_2 \leq \|\mathcal{K}_\sigma\|_2 + \varepsilon_{n,\sigma}$ and [theorem 3.2](#). □

Corollary 5.2 (Power-law resolution thresholds). *Suppose $n(\sigma) \asymp \sigma^{-p}$. Then the bounds above have leading powers*

$$\varepsilon_{n,\sigma} = O(\sigma^{p-3/2}), \quad (24)$$

$$\delta_{n,\sigma} = O(\sigma^{p-2}) + O(\sigma^{2p-3}). \quad (25)$$

Consequently

$$n(\sigma) \sigma^{3/2} \rightarrow \infty \implies G_{n,\sigma} - \mathcal{K}_\sigma \rightarrow 0 \text{ in } \mathfrak{S}_2, \quad (26)$$

$$n(\sigma) \sigma^2 \rightarrow \infty \implies G_{n,\sigma}^2 - \mathcal{K}_\sigma^2 \rightarrow 0 \text{ in } \mathfrak{S}_1. \quad (27)$$

For pure powers, $p > 3/2$ and $p > 2$, respectively, are sufficient. At $p = 3/2$ or $p = 2$, the corresponding leading bound remains order one.

This explains why an engineering schedule $n\sigma \simeq 20.48$, adequate for visually resolving a Gaussian row and used in the earlier cloud audits, does not certify an absolute Hilbert–Schmidt small-noise limit. It was never designed for that topology.

5.1 Midpoint, row normalization, and cutoff

The same exponents extend asymptotically to the exact-real midpoint family. Gaussian differentiation and (8) give, for every fixed derivative order,

$$\left\| \partial_x^a \partial_y^b k_\sigma \right\|_{L^2} = O\left(\sigma^{-a-b-1/2}\right). \quad (28)$$

The midpoint Peano estimate therefore has size

$$O(n^{-2}\sigma^{-5/2}) + O(n^{-4}\sigma^{-9/2}), \quad (29)$$

and the exact row-normalization correction has size $O(n^{-2}\sigma^{-5/2})$. Relative to the leading Galerkin error $O(n^{-1}\sigma^{-3/2})$, these have ratios $O((n\sigma)^{-1})$ and $O((n\sigma)^{-3})$. Thus they are lower order under (26).

The adaptive Gaussian cutoff from RH-39 has an explicit Frobenius-square bound. With $L_n = \max\{8, 2\sqrt{\log n}\}$, it is also lower order whenever $n\sigma \rightarrow \infty$. The displayed constants in this paper remain those of the canonical cell-average lift; the midpoint statement here is an asymptotic scaling corollary, not a new interval certificate at every noise width.

5.2 Determinant disks

For

$$D_{n,\sigma}^K(w) = \det(I - wG_{n,\sigma}^2), \quad D_\sigma^K(w) = \det(I - w\mathcal{K}_\sigma^2),$$

the standard determinant continuity bound gives

$$\begin{aligned} \sup_{|w| \leq R} |D_{n,\sigma}^K(w) - D_\sigma^K(w)| &\leq R\delta_{n,\sigma} \\ &\times \exp\left[1 + RA_0^2\sigma^{-1} + R(A_0\sigma^{-1/2} + \varepsilon_{n,\sigma})^2\right]. \end{aligned} \quad (30)$$

If $R_\sigma = \rho\sigma$ and $n\sigma \rightarrow \infty$, the exponential remains bounded and

$$R_\sigma\delta_{n,\sigma} = O((n\sigma)^{-1}) + O((n\sigma)^{-2}) \rightarrow 0. \quad (31)$$

Thus shrinking disks require only $p > 1$ in this sufficient bound. For fixed $R > 0$, the exponential contains $e^{2RA_0^2/\sigma}$. Making (30) tend to zero by this generic inequality requires $n\sigma^2$ to dominate an exponential in $1/\sigma$.

Remark 5.3 (A sufficient wall, not a complexity lower bound). The fixed-disk exponential is the price of a global determinant Lipschitz estimate using only trace norms. It does not prove that every determinant algorithm requires exponentially many grid points. It shows that a sharper fixed-disk theorem must exploit spectral factorization or cloud renormalization rather than only the raw trace-ideal norm.

6 The missing uniform peripheral theorem

For every sufficiently small σ , strong positivity gives the Perron branch and small-noise spectral stability gives a simple real negative branch with $\lambda_-(\sigma) \rightarrow -1$ [2]. Thus the intrinsic bulk

$$\mathcal{B}_\sigma = \mathcal{K}_\sigma - \mathcal{Q}_{+,\sigma} - \mathcal{Q}_{-,\sigma} \quad (32)$$

exists analytically. What is not yet available is a quantitative L^2 contour theory uniform as $\sigma \downarrow 0$.

Condition 6.1 (Uniform peripheral transport). There are exponents $q, r \geq 0$ and constants independent of small σ such that

$$\|\mathcal{Q}_{+, \sigma} + \mathcal{Q}_{-, \sigma}\|_{\mathfrak{S}_2} = O(\sigma^{-q}), \quad (33)$$

$$\|Q_{+, n, \sigma} + Q_{-, n, \sigma} - \mathcal{Q}_{+, \sigma} - \mathcal{Q}_{-, \sigma}\|_{\mathfrak{S}_2} = O(n^{-1}\sigma^{-r}). \quad (34)$$

Theorem 6.2 (Conditional bulk mesh law). *Under condition 6.1, put*

$$a = \max\{1/2, q\}, \quad b = \max\{3/2, r\}. \quad (35)$$

Then

$$\|\mathcal{B}_\sigma\|_{\mathfrak{S}_2} = O(\sigma^{-a}), \quad (36)$$

$$\|B_{n, \sigma} - \mathcal{B}_\sigma\|_{\mathfrak{S}_2} = O(n^{-1}\sigma^{-b}), \quad (37)$$

$$\|B_{n, \sigma}^2 - \mathcal{B}_\sigma^2\|_{\mathfrak{S}_1} = O(n^{-1}\sigma^{-a-b}) + O(n^{-2}\sigma^{-2b}). \quad (38)$$

In particular,

$$n(\sigma)\sigma^{a+b} \rightarrow \infty \quad (39)$$

is sufficient for bulk-square trace-norm convergence.

The matched Markov exponents $q = 1/2$, $r = 3/2$ reproduce the power $a + b = 2$. If moving contours have $O(1)$ geometry and their continuum and discrete resolvents are $O(\sigma^{-\beta})$, a direct resolvent-identity route gives the coarse assignments

$$q = \beta, \quad r = \frac{3}{2} + 2\beta, \quad (40)$$

and hence the sufficient mesh power

$$p_\beta = \max\{1/2, \beta\} + \frac{3}{2} + 2\beta. \quad (41)$$

Thus $\beta = 0$ gives $p_0 = 2$, while $\beta = 1/2$ already gives $p_{1/2} = 3$. Determining the actual resolvent-growth exponent is the next strict contour problem.

RH-45 proves condition 6.1 at the single width $\sigma = 10^{-2}$ with explicit constants. This paper does not extrapolate that certificate to smaller noise.

7 The deterministic two-step double pole

For fixed $\sigma > 0$, define the one-step regularized intrinsic determinant

$$D_{\sigma, \text{bulk}, 2}(z) = \det_2(\mathbf{I} - z\mathcal{B}_\sigma). \quad (42)$$

Its symmetric product is the strict two-step determinant

$$\mathcal{F}_\sigma(w) = \det(\mathbf{I} - w\mathcal{B}_\sigma^2), \quad \mathcal{F}_\sigma(z^2) = D_{\sigma, \text{bulk}, 2}(z)D_{\sigma, \text{bulk}, 2}(-z). \quad (43)$$

Every $\mathcal{F}_\sigma$ is entire.

The deterministic parity-centered one-step cycle germ has already been identified exactly [3]:

$$\hat{D}_{0, \text{bulk}, 2}(z) = \frac{\mathcal{G}(z)}{1 - z^2/\lambda}, \quad \lambda = 1.678573510428322\dots, \quad (44)$$

where $\mathcal{G}$ is holomorphic and nonzero for $|z| < \lambda$.

Theorem 7.1 (Genuine two-step double pole). *There is a function $\mathcal{H}$, holomorphic and nonzero for $|w| < \lambda^2$, such that the deterministic two-step germ is*

$$\widehat{\mathcal{F}}_0(w) = \frac{\mathcal{H}(w)}{(1 - w/\lambda)^2}. \quad (45)$$

Consequently $w = \lambda$ is an actual double pole and the Taylor series of $\widehat{\mathcal{F}}_0$ has exact radius λ .

Proof. The product $\mathcal{G}(z)\mathcal{G}(-z)$ is even and holomorphic for $|z| < \lambda$, so it descends through $w = z^2$ to a holomorphic function $\mathcal{H}(w)$ for $|w| < \lambda^2$. It is nonzero there because both factors are nonzero. Multiplying (44) at z and $-z$ gives (45). Since $\sqrt{\lambda} < \lambda$, $\mathcal{H}(\lambda) = \mathcal{G}(\sqrt{\lambda})\mathcal{G}(-\sqrt{\lambda}) \neq 0$, so the pole is not canceled. $\square$

Fixed-length Gaussian localization and the square-root parity law give coefficientwise convergence of the one-step determinants [2, 3, 5]. Taking the symmetric product yields the corresponding two-step statement.

Proposition 7.2 (Two-step coefficient bridge). *Every fixed Taylor coefficient of $\mathcal{F}_\sigma(w)$ converges, as $\sigma \downarrow 0$, to the corresponding coefficient of $\widehat{\mathcal{F}}_0(w)$.*

Theorem 7.3 (Two-step normal-family obstruction). *Let $R > \lambda$. No sequence $\sigma_j \downarrow 0$ can make $\mathcal{F}_{\sigma_j}$ converge locally uniformly to a finite holomorphic function on $|w| < R$. For every $\sigma_0 > 0$, the family*

$$\{\mathcal{F}_\sigma : 0 < \sigma < \sigma_0\} \quad (46)$$

is not locally bounded on that disk.

Proof. If a locally uniform limit F existed, convergence of derivatives at zero and proposition 7.2 would identify its Taylor germ with $\widehat{\mathcal{F}}_0$. The identity theorem would then continue that equality toward $w = \lambda$. But F is holomorphic there, whereas theorem 7.1 makes $\widehat{\mathcal{F}}_0$ unbounded in every punctured neighborhood. If the full family were locally bounded, Montel's theorem would provide a locally convergent subsequence from every $\sigma_j \downarrow 0$, giving the same contradiction [1]. $\square$

This obstruction survives perfect spatial resolution. The mesh problem and the pole problem are therefore independent: one must solve both, and solving only the first cannot produce an entire small-noise limit.

8 Canonical squared-cloud resolution

Let

$$\Pi_N(q) = 1 + q + \cdots + q^N = \frac{1 - q^{N+1}}{1 - q}. \quad (47)$$

The canonical one-step cloud resolving the simple pole consists of

$$\mu_{N,k}^\pm = \lambda^{-1/2} e^{\pm i k \pi / (N+1)}, \quad 1 \leq k \leq N. \quad (48)$$

Theorem 8.1 (Squared geometric pole resolution). *The two-step determinant over the $2N$ values in (48) is*

$$\prod_{k=1}^N \prod_{\epsilon \in \{+, -\}} \left(1 - w(\mu_{N,k}^\epsilon)^2\right) = \Pi_N(w/\lambda)^2. \quad (49)$$

Every nontrivial $(N+1)$ st-root zero therefore has multiplicity two in the w -plane. Moreover, locally uniformly for $s \in \mathbb{C}$,

$$\frac{\Pi_N(e^{s/(N+1)})^2}{(N+1)^2} \longrightarrow \left(\frac{e^s - 1}{s}\right)^2, \quad (50)$$

with the removable value one at $s = 0$.

Proof. The one-step identity is $\prod(1 - z\mu_{N,k}^\epsilon) = \Pi_N(z^2/\lambda)$. Multiply it at z and $-z$, or directly square the cloud eigenvalues, to obtain (49). Inserting $q = e^{s/(N+1)}$ in (47) gives

$$\frac{\Pi_N(q)}{N+1} = \frac{e^s - 1}{(N+1)(e^{s/(N+1)} - 1)} \rightarrow \frac{e^s - 1}{s}.$$

Squaring proves (50). $\square$

The endpoint Gaussian resolution theorem gives the independent rank clock

$$N_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1) \quad (51)$$

for the canonical endpoint resolution operator [4]. Identifying this singular-value rank with the actual number of noisy bulk resonances remains conjectural. The squared finite section is exact once an ideal degree N is chosen; the connection $N = N_\sigma$ for the full Markov operator is not claimed as a theorem here.

9 Numerical audits

The calculations have two purposes: verify the sharp row constant and reprocess the existing one-step clouds in the intrinsic two-step variable. Neither computation is used in the double-pole theorem.

9.1 Power schedules

For display, set

$$n_p(\sigma) = \lceil 65536(0.01/\sigma)^p \rceil. \quad (52)$$

All dimensions in table 1 are theoretical evaluations of the closed-form bounds; matrices with billions of rows were not built.

p	$n_p(10^{-4})$	HS upper	square trace upper	$\log_{10} \Delta_{R=\sigma}$	$\log_{10} \Delta_{R=0.1}$
1.00	6.5536×10^6	5.5242×10^{-1}	2.3503×10^2	2.7360	3932.40
1.50	6.5536×10^7	5.5242×10^{-2}	2.3476×10^1	1.7263	3922.22
2.00	6.5536×10^8	5.5242×10^{-3}	2.3473	0.7254	3920.30
2.25	2.0724×10^9	1.7469×10^{-3}	7.4227×10^{-1}	0.2253	3919.73
2.50	6.5536×10^9	5.5242×10^{-4}	2.3473×10^{-1}	-0.2747	3919.21

Table 1: Uniform analytic bounds at $\sigma = 10^{-4}$. The determinant columns use the standard trace-ideal continuity estimate. The fixed-disk numbers illustrate its exponential wall; they are not predictions of actual floating determinant errors.

9.2 Squared archived clouds

For each RH-15 cloud, let r_σ be its arithmetic mean radius and define

$$C_\sigma^{(2)}(w) = \prod_{\mu \in \mathcal{C}_\sigma} (1 - w\mu^2). \quad (53)$$

We compare

$$\frac{C_\sigma^{(2)}(r_\sigma^{-2} e^{s/(N_\sigma+1)})}{C_\sigma^{(2)}(r_\sigma^{-2})} \quad (54)$$

with the finite section $[\Pi_{N_\sigma}(e^{s/(N_\sigma+1)})/(N_\sigma+1)]^2$.

At $\sigma = 10^{-4}$, the selected one-step cloud has 14 eigenvalues and $N_\sigma = 7$. Its mean radius is 0.7302687845, while the corresponding two-step edge center is $r_\sigma^{-2} = 1.8751435740$. On the seven-point set

$$\{-2, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, 2\},$$

the mean observed-to-finite-section error is 0.1274. Restricting to $|s| \leq 1$, the mean is 0.03633 and the maximum is 0.11590. The data are compatible with the squared finite-section mechanism, but the error is not monotone across the seven archived noise levels. We therefore report a diagnostic, not a convergence fit.

Small-noise mesh laws and the bulk two-step double-pole obstruction

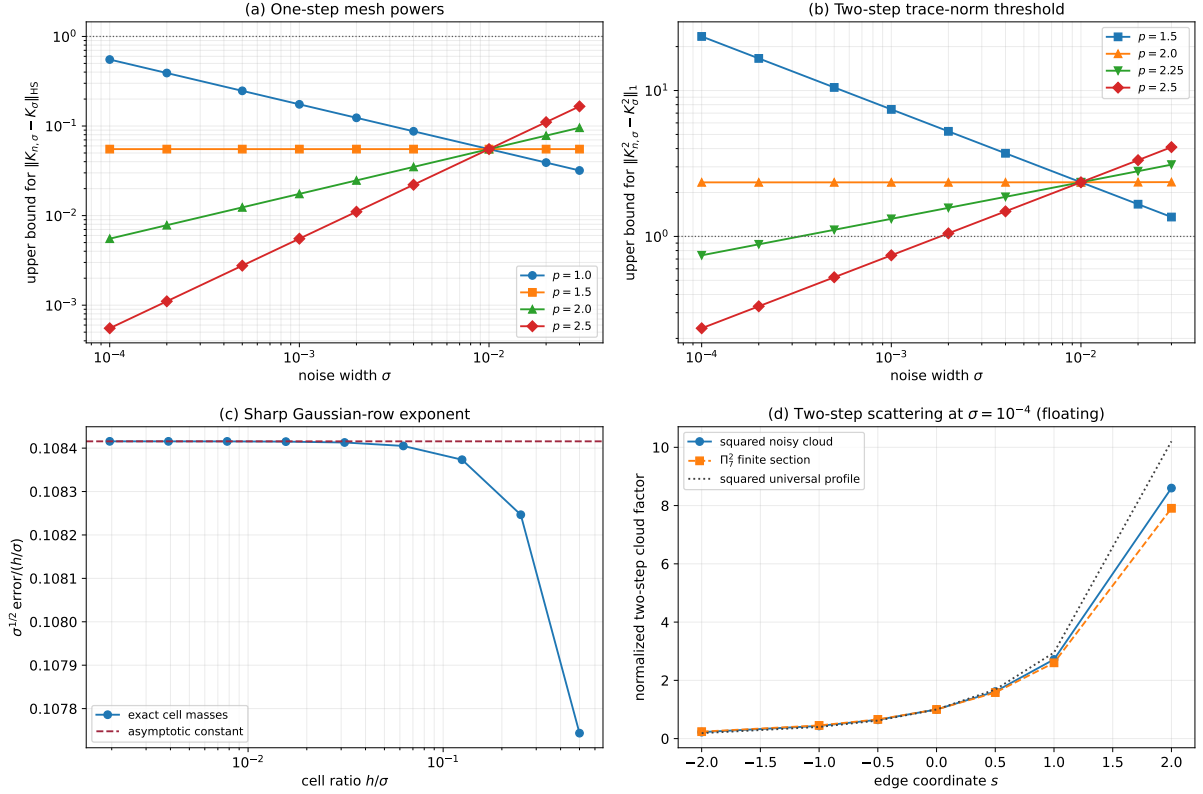


Figure 1: Small-noise mesh and pole-resolution summary. (a) Explicit Hilbert–Schmidt mesh bounds for $n_p(\sigma) \asymp \sigma^{-p}$; $p = 3/2$ is critical. (b) Bulk-free Markov square trace-norm bounds; $p = 2$ is critical and $p > 2$ decays. (c) Exact normalized Gaussian cell masses recover the sharp constant in (18). (d) The archived $\sigma = 10^{-4}$ resonance cloud after squaring, compared with the exact Π_7^2 finite section and its universal large- N profile. Panel (d) is floating evidence only.

10 What is closed and what remains

The maze now has two separately marked walls.

1. **Spatial wall.** Absolute Hilbert–Schmidt approximation of a narrowing Gaussian kernel requires more than fixed points per noise width. The canonical sufficient scales are $n\sigma^{3/2} \rightarrow \infty$ at one step and $n\sigma^2 \rightarrow \infty$ for trace-norm squares.
2. **Analytic wall.** Even perfect continuum resolution cannot yield an unrenormalized entire small-noise two-step determinant. The coefficient germ has a genuine double pole at $w = \lambda$.

The next positive gate is correspondingly precise: prove a moving-contour weighted-Riesz theorem strong enough to determine the exponents q, r in [condition 6.1](#). A uniform resolvent bound would retain the $p > 2$ two-step mesh law. Polynomial resolvent growth would quantify exactly how much faster the grid must grow.

Beyond that contour gate, the determinant must be renormalized by an actual noisy cloud factor. The canonical candidate is

$$\Pi_{N_\sigma}(w/\lambda)^2, \quad (55)$$

with the rank clock [\(51\)](#). Proving that division by the actual cloud leaves a normal residual family would convert the negative double-pole theorem into a positive scattering continuation.

None of these statements identifies a trace with primes or prime powers, constructs a self-adjoint operator, derives a $T \log T$ counting law, or locates a Riemann zero. The result is a resolution and complex-analytic theorem for one explicit noisy dynamical family.

Data and code availability

All source code, certificates, floating pilots, tests, figures, hashes, and the manuscript are available in https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-46-small-noise-mesh-double-pole.

References

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